CROCODALE

An auto-scrolling platformer game for mobile devices

DESCRIPTION

Maintenance document to familiarize the reader with the CrocoDale -mobile game. The game features 18 colorful and fun-packed levels with various enemies, obstacles and collectibles. And most importantly, it has cosmetic hats and cute animals.

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Master's Project

1.0 Introduction

This document presents the CrocoDale -mobile game. The game is created using Unity game engine, and the genre of the game is auto scrolling platformer. This document aims to introduce a developer who is unfamiliar with the game so, that they could continue developing it and add new features or fix problems with existing ones.

1.1 Structure of the document

The game and its features are presented more extensively in the chapter 2. Chapter 3 undergoes the file structure and technical implementation in Unity. In chapter 4, the main design principles of the game are presented, as well as the unused assets and concepts. Chapter 5 discusses the launch and the future of the game.

2.0 CrocoDale

In CrocoDale the player controls a young crocodile through 18 levels. In order to get to the goal of each level, the player has to time their jumps correctly. There are also additional collectibles to be collected.

2.1 Gameplay Loop

The core gameplay loop of the game is simple. When starting a level, the player character starts to move to the right automatically. When touching the screen, the character jumps, the jump height depending on the duration the screen is held down. The player has to avoid pitfalls and enemies by jumping, while also collecting optional collectibles throughout the level. The levels take approximately half a minute to complete, if the player doesn’t have to start over at any point. By collecting each collectible from one world, the player can purchase a cosmetic hat from the store.

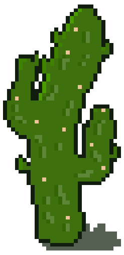
2.2 Worlds and levels

The game is broken down to 3 worlds, each containing 6 levels. Each world has unique graphics, collectibles, enemies and gimmicks. For example, the jungle -world has giant leaves that quickly start to fall as the player touches them, creating a unique feeling when compared to other worlds.

2.3 Enemies and obstacles

The player has no way of destroying enemies, so they can only be avoided. The player doesn’t have health bar either, so upon touching any enemy the level has to be started over. There are different types of enemies and obstacles in every world, explained below:

2.3.1 Savannah



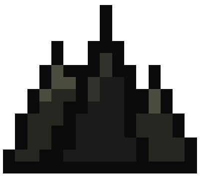
Cactus – Doesn't do anything, the player just has to jump over it.

Snake – Starts moving to the left when the player comes near. Player has to jump over it.

Mosquito – Starts to home towards the player when they come near. Player has to time their jump correctly just before the impact to avoid it.¨

2.3.2 Cave

Bat - Starts to home towards the player when they come near. Player has to time their jump correctly just before the impact to avoid it. There's also a variant that flies straight forward.

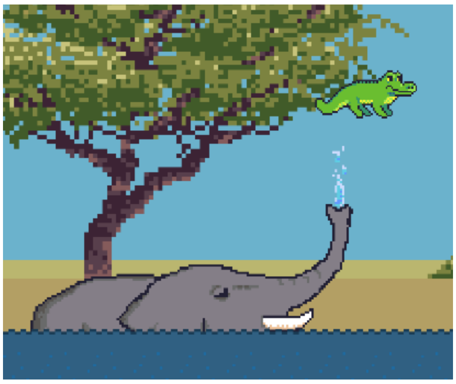


Spikes – The player has to jump over to avoid them.

2.3.3 Jungle

Monkey – Rolls a coconut towards the player, and escapes with a vine. Player has to jump over the coconut to avoid it, or in some cases wait in a safe spot to let it pass over.

2.4 Gimmicks



Elephant – The savannah has elephants that launch the player with predetermined force.

The cave has no special gimmick, but **some of the levels get darker** as the player progresses.

Giant leaves – The jungle has giant leaves that start to fall when the player touches them

2.5 Music and sound effects

Music in the game is upbeat and cheerful. It comes mostly from *8-bit Action Free* [1] -package by Moppysound from the Unity Asset store. There's also a single music track from the *8-bit 8 Pack Music Loops* [2] by Fancy Lizard Studio.

The sound effects used in the game are from *(Free) Retro SFX Pack* [3] from 3rd Note Inc and *Universal Sound FX* [4] by Imphenzia.

2.6 Story and narrative

CrocoDale is very simple in terms of story and narrative. Starting screen of the game contains a short animation where a group of mischievous monkeys steal a bunch of eggs, leaving one behind. Shortly after, the player character crocodile hatches from the remaining egg, and goes after the monkeys in order to save their yet-to-be-hatched siblings. At the end of every level, there is an egg that the player crocodile takes along.

3.0 Implementation

3.1 Unity version

Originally the game was implemented in Unity version 2017.3. However, because Google Play store no longer supports .apk files, but requires .aab files, the version had to be updated before release to 2017.4 as it had the support for building .aab files. Updating to the newer Unity versions (2018.x+) would bring some new useful features, such as 2D lighting. The tilemap -feature which is used to create the levels was still in beta in 2017.4, so that would probably also be better in the newer versions. However, when updating the major Unity version some things stop working and require rework, such as the beforementioned tilemaps. For the convenience, development has been done in the 2017.4 version and no updating is recommended, unless absolutely necessary.

3.2 File structure

The main folder is named Krokotiilipeli, which contains the files required by Unity engine, as well as the Assets-folder. Assets-folder contains pretty much everything the developer needs to be concerned with. It contains the Unity-scenes and fonts required by the game, as well as the following subfolders:

3rd Note SFX Pack_Vol.1 Wonder FX

8-bit 8Pack

8-Bit_Action_Free

Animations

Prefab

Scripts

Sprites

Tilemap

Universal Sound FX

3.3. Gameplay Scripts

Most scripts and their functions are apparent from the name of the script file, for example CameraController controls the camera, CoconutMonkey contains the functionality of the coconut monkey enemy, etc.

PlayerController has the most functionality, even some things that are not directly related to controlling the player. It is used to read the player inputs and control the player and its collisions. It also controls the particles and plays the sound effects of the player character as well as the main level music. It also used to save completed levels and collected collectibles in the GlobalObject -script, which contains things like saving and loading the player data.

GlobalObject -object is initialized in the level select -scene. It loads player data from PlayerPrefs -file, which is then used to determine which levels are completed, and how many collectibles are found from each one. It also saved which of the hats, if any, the player has equipped from the shop.

Prefabs -folder contains premade objects, such as the player character, enemies and UI-elements, which are reused thorough the game. Some of these objects have variables that can be modified from the Unity editor, for example the force which the elephant hurls the player upwards and forwards.

3.4 User Interface

The user interface of the game is done by using the default Unity UI -components. Despite its name, StartScreen -script contains most functions used by the user interface, even in level select and in levels. This includes functions like choosing and restarting the level, changing music and effect volumes as well as switching to the loading screen when starting a new level.

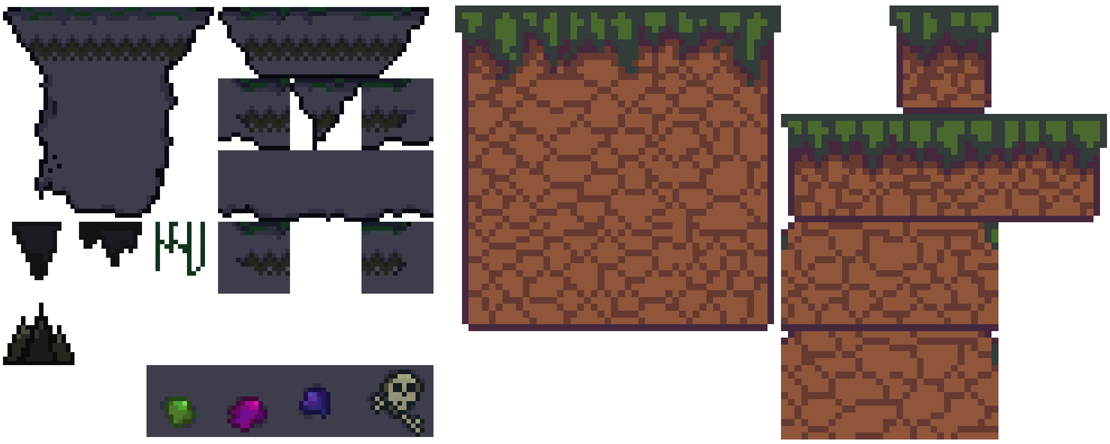
Screenshot of the level select menu (there's a cool effect of the bananas moving in the background, done with the particle system, which you cannot see from the screenshot)

3.5 Sprites

All sprites used in the game are hand-drawn and animated with Asesprite-software. Not many different colors are used in a sprite, and for example shadowing is done with only single a bit darker color, if at all. The aim is to create a colorful and simple look for the game, avoiding too complex sprites, as it the goal is to keep it simple, so objects are well distinguishable from the small mobile screen.

3.6 Levels

The levels are made up with tilemaps, which contain collisions for the ground. The levels are then populated with enemies and hazard. Finally, they are decorated with backgrounds and other visual sprites. The order of layers is important here: background layer is rendered with a different camera which is farther away than the main camera, making the background move slower than the main character which follows the player. The layers are also important for collisions, for example "death" layer instantly ends the run when player character touches it.



Cave and jungle tilemaps.

4.0 Design

"Video games are just meant for one thing: Fun! Fun for everyone."

-Satoru Iwata

4.1 Principles

The main design principle for CrocoDale is to keep it simple and have fun. If developing the game is fun for the developer(s), it should also reflect to the player.

4.1.1 Revenue

The game contains no advertisement or paid content, as it would steer the game to less fun and more business-like direction. In case some way to support the developer(s) wants to be added to the game, cosmetic hat(s) with a low price could be added.

4.1.2 Metrics

Currently metrics aren't used at all, but in case they are implemented to the game, they should only be used to fix critical flaws (such as too difficult levels), but not to optimize Daily Active Users, Average Session Length etc. as it would steer the game to a more metric driven direction.

4.1.3 Gameplay

As for the gameplay, it should remain relatively simple, so people of all ages and skill levels can enjoy playing it. As for the difficulty, the game shouldn't be too hard, but the levels should take at least few tries to complete, and a few more if all collectibles are collected.

4.2 Unused assets and concepts

The game contains a few assets that were unused. During the development there were also a few ideas that could be implemented at a later time.

4.2.1 Eyeglass Snake

At some point, the game was to feature a wise-looking snake that would teach the player the basic controls. However, this idea was scrapped, but the sprite for the snake still remains in the game files. It could be re-used to create a more difficult variant of the snake enemy, for example one that doesn't immediately start to move when the player approaches but instead waits a bit and then sprints at double speed to make jumping over it harder.

4.2.1 Quicksand

The savannah world was to feature quicksand-pits that the player could fall into. The player would then have to smash jumping button rapidly to escape it. However, there's no quicksand in the savannahs and the idea was scrapped. In case a desert-world is added to the game at a later stage, the unused sprites and scripts for the quicksand still exists in the game files.

5.0 Future

Beta version of the game is released in Google Play store. It contains the features presented in this document. With this version, the aim is to easily share the game with family and friends, and to balance levels better and to find and fix bugs according to their feedback. It's also more professional to tell people to just search "CrocoDale" from the Google Play store, than it is to hassle around with Unity beta build .apk-files. As a developer it's also a stepping stone to release an actual game for the public, simple as it may be, on the Google Play store for the first time. A version for iOS will be made and released on Apple Store when there's access to iOS -device to test it.

There are also plans to add more levels and content in case there is demand. For example, Space -world with a space helmet hat and Arctic -world with a furry hat could be added, as well as longer special holiday levels that reward cosmetic hats such as Santa Claus hat or Halloween pumpkin hat.

References

[1] https://assetstore.unity.com/packages/audio/music/electronic/8-bit-action-free-19827

[2] https://assetstore.unity.com/packages/audio/music/electronic/8-bit-8-pack-music-loops-60232

[3] https://assetstore.unity.com/packages/audio/sound-fx/free-retro-sfx-pack-43256

[4] https://assetstore.unity.com/packages/audio/sound-fx/universal-sound-fx-17256